KAIZEN MATHS

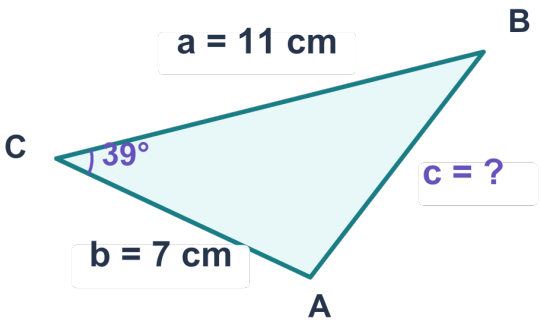
Lesson starter

Sine Rule, Cosine Rule and Triangle Area

Cosine Rule & Mixed

Name _____

SA1



In triangle ABC, a = 11 cm, b = 7 cm, and included angle C = 39°. Find c to 2 d.p.

ALG-2B2B-L2 · SA

Show your working

KAIZEN MATHS

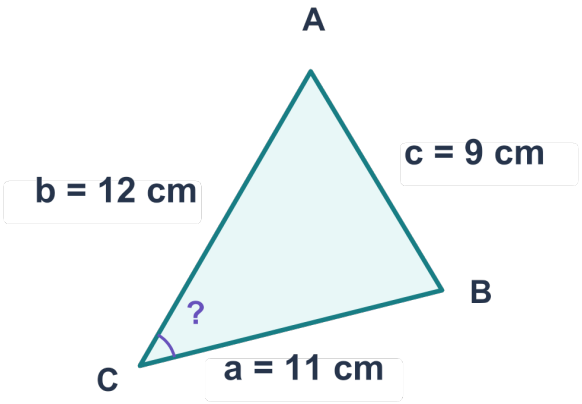
Lesson starter

Sine Rule, Cosine Rule and Triangle Area

Cosine Rule & Mixed

Name _____

SB1



In triangle ABC, a = 11 cm, b = 12 cm, and c = 9 cm. Find angle C to 1 d.p.

ALG-2B2B-L2 · SB

Show your working

KAIZEN MATHS

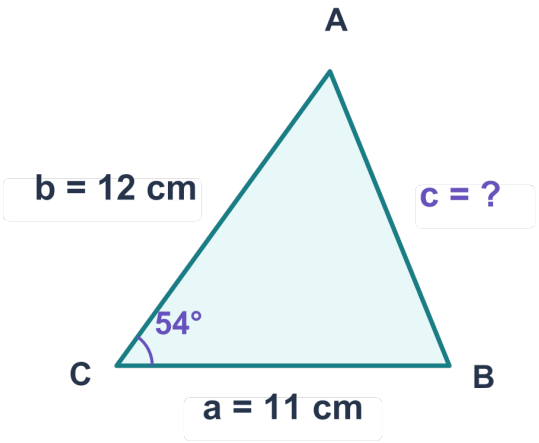
Lesson starter

Sine Rule, Cosine Rule and Triangle Area

Cosine Rule & Mixed

Name _____

SC1



In triangle ABC, a = 11 cm, b = 12 cm, and included angle C = 54°. Find c to 2 d.p.

ALG-2B2B-L2 · SC

Show your working

KAIZEN MATHS

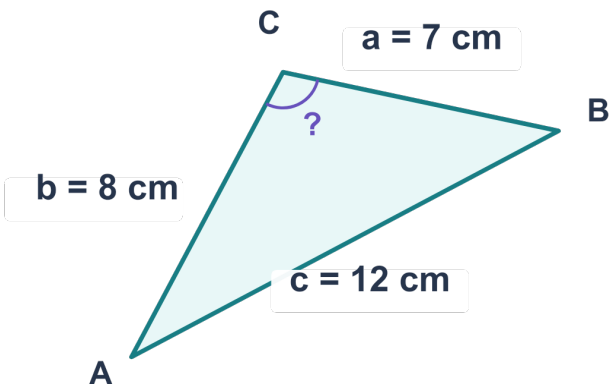
Lesson starter

Sine Rule, Cosine Rule and Triangle Area

Cosine Rule & Mixed

Name _____

SD1



In triangle ABC, a = 7 cm, b = 8 cm, and c = 12 cm. Find angle C to 1 d.p.

ALG-2B2B-L2 · SD

Show your working

KAIZEN MATHS

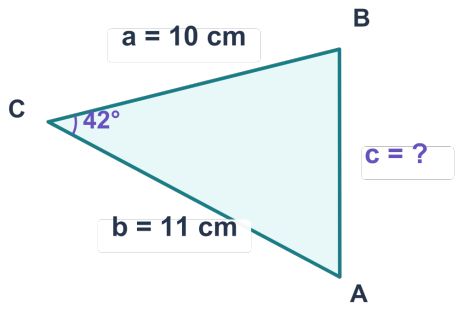
Exit ticket

Sine Rule, Cosine Rule and Triangle Area

Cosine Rule & Mixed

Name _____

EA1



In triangle ABC, a = 10 cm, b = 11 cm, and included angle C = 42°. Find c to 2 d.p.

Show your working

KAIZEN MATHS

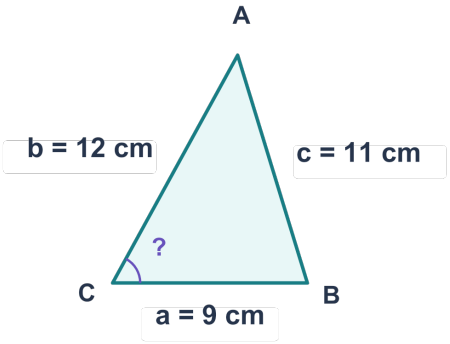
Exit ticket

Sine Rule, Cosine Rule and Triangle Area

Cosine Rule & Mixed

Name _____

EB1



In triangle ABC, a = 9 cm, b = 12 cm, and c = 11 cm. Find angle C to 1 d.p.

Show your working

KAIZEN MATHS

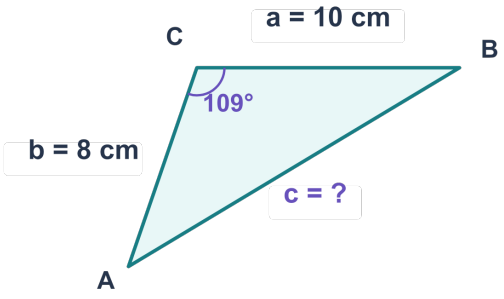
Exit ticket

Sine Rule, Cosine Rule and Triangle Area

Cosine Rule & Mixed

Name _____

EC1



In triangle ABC, a = 10 cm, b = 8 cm, and included angle C = 109°. Find c to 2 d.p.

Show your working

KAIZEN MATHS

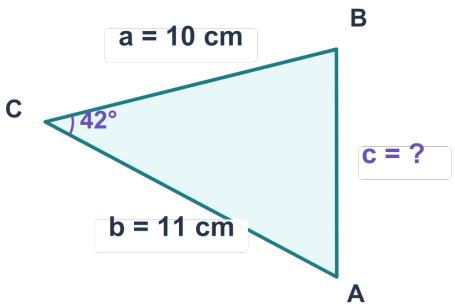
Exit ticket

Sine Rule, Cosine Rule and Triangle Area

Cosine Rule & Mixed

Name _____

EA1



In triangle ABC, a = 10 cm, b = 11 cm, and included angle C = 42°. Find c to 2 d.p.

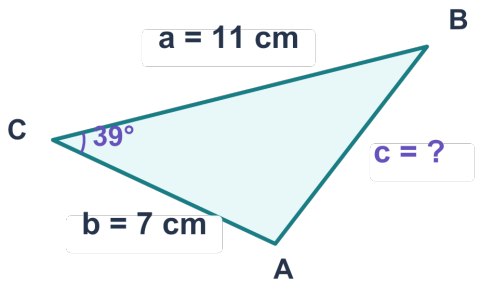
Show your working

Teacher answers & working

Sine Rule, Cosine Rule and Triangle Area · Cosine Rule & Mixed

ALG-2B2B-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1



In triangle ABC, a = 11 cm, b = 7 cm, and included angle C = 39°. Find c to 2 d.p.

Answer:

$c = 7.09$ cm

Working:

Cosine Rule: $c^2 = a^2 + b^2 - 2ab \cos C$

$c^2 = 11^2 + 7^2 - 2 \times 11 \times 7 \times \cos 39°$

$\cos 39° \approx 0.7771$

$c^2 \approx 50.3195$

$c \approx \sqrt{50.3195} \approx 7.0936$

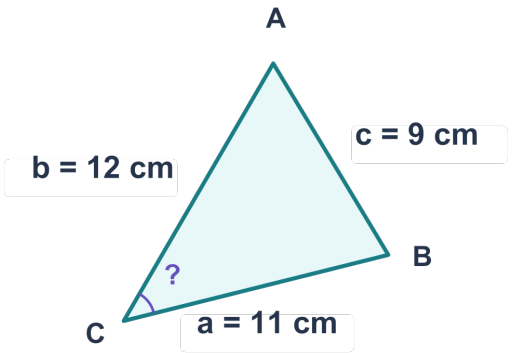
$c = 7.09$ cm

Teacher answers & working

Sine Rule, Cosine Rule and Triangle Area · Cosine Rule & Mixed

ALG-2B2B-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SB1



In triangle ABC, a = 11 cm, b = 12 cm, and c = 9 cm. Find angle C to 1 d.p.

Answer:

$C = 45.8°$

Working:

$\cos C = (a^2 + b^2 - c^2) / (2ab)$

$\cos C = (121 + 144 - 81) / (2 \times 11 \times 12)$

$\cos C \approx 0.6970$

$C \approx \cos^{-1}(0.6970) \approx 45.8°$

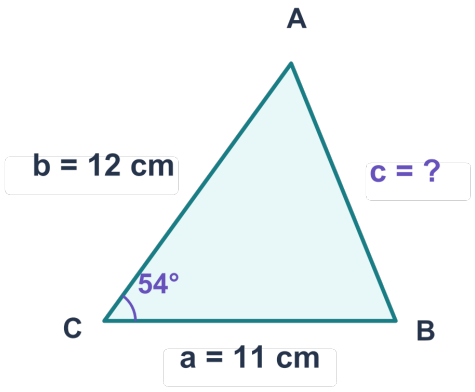
$C = 45.8°$

Teacher answers & working

Sine Rule, Cosine Rule and Triangle Area · Cosine Rule & Mixed

ALG-2B2B-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SC1



In triangle ABC, a = 11 cm, b = 12 cm, and included angle C = 54°. Find c to 2 d.p.

Answer:

$c = 10.48$ cm

Working:

Cosine Rule: $c^2 = a^2 + b^2 - 2ab \cos C$

$c^2 = 11^2 + 12^2 - 2 \times 11 \times 12 \times \cos 54°$

$\cos 54° \approx 0.5878$

$c^2 \approx 109.8247$

$c \approx \sqrt{109.8247} \approx 10.4797$

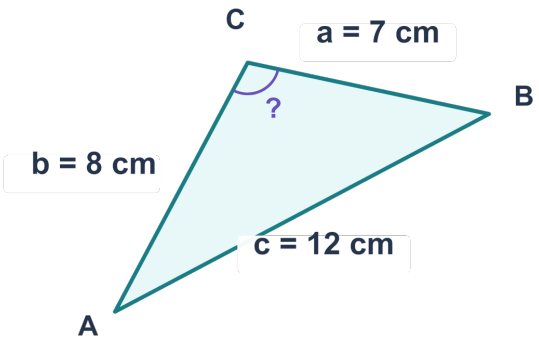
$c = 10.48$ cm

Teacher answers & working

Sine Rule, Cosine Rule and Triangle Area · Cosine Rule & Mixed

ALG-2B2B-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SD1



In triangle ABC, a = 7 cm, b = 8 cm, and c = 12 cm. Find angle C to 1 d.p.

Answer:

$C = 106.1°$

Working:

$\cos C = (a^2 + b^2 - c^2) / (2ab)$

$\cos C = (49 + 64 - 144) / (2 \times 7 \times 8)$

$\cos C \approx -0.2768$

$C \approx \cos^{-1}(-0.2768) \approx 106.1°$

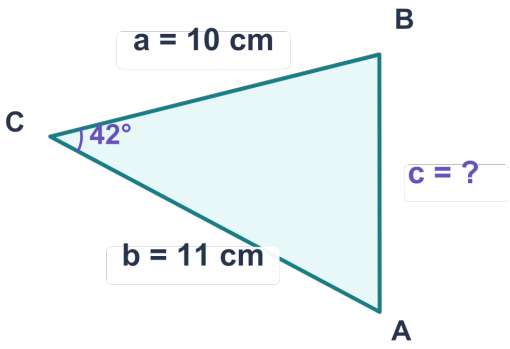
$C = 106.1°$

Teacher answers & working

Sine Rule, Cosine Rule and Triangle Area · Cosine Rule & Mixed

ALG-2B2B-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1



In triangle ABC, a = 10 cm, b = 11 cm, and included angle C = 42°. Find c to 2 d.p.

Answer:

$c = 7.58$ cm

Working:

Cosine Rule: $c^2 = a^2 + b^2 - 2ab \cos C$

$c^2 = 10^2 + 11^2 - 2 \times 10 \times 11 \times \cos 42°$

$\cos 42° \approx 0.7431$

$c^2 \approx 57.5081$

$c \approx \sqrt{57.5081} \approx 7.5834$

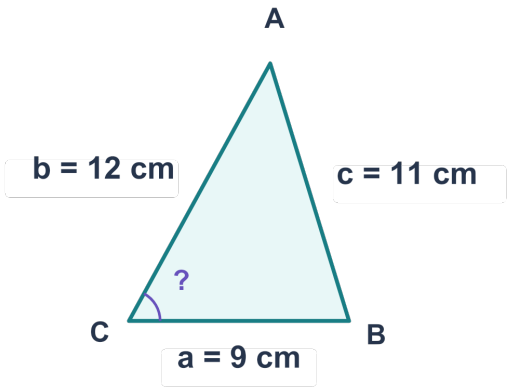
$c = 7.58$ cm

Teacher answers & working

Sine Rule, Cosine Rule and Triangle Area · Cosine Rule & Mixed

ALG-2B2B-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EB1



In triangle ABC, a = 9 cm, b = 12 cm, and c = 11 cm. Find angle C to 1 d.p.

Answer:

$C = 61.2°$

Working:

$\cos C = (a^2 + b^2 - c^2) / (2ab)$

$\cos C = (81 + 144 - 121) / (2 \times 9 \times 12)$

$\cos C \approx 0.4815$

$C \approx \cos^{-1}(0.4815) \approx 61.2°$

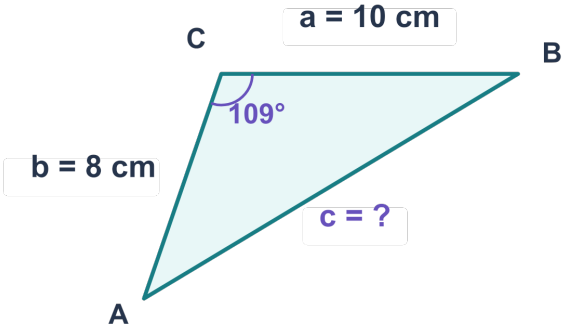
$C = 61.2°$

Teacher answers & working

Sine Rule, Cosine Rule and Triangle Area · Cosine Rule & Mixed

ALG-2B2B-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EC1



In triangle ABC, a = 10 cm, b = 8 cm, and included angle C = 109°. Find c to 2 d.p.

Answer:

$c = 14.70$ cm

Working:

Cosine Rule: $c^2 = a^2 + b^2 - 2ab \cos C$

$c^2 = 10^2 + 8^2 - 2 \times 10 \times 8 \times \cos 109°$

$\cos 109° \approx -0.3256$

$c^2 \approx 216.0909$

$c \approx \sqrt{216.0909} \approx 14.7000$

$c = 14.70$ cm